Cross-Category Generalization in Bengali Fake News Detection: A Leave-One-Category-Out Study

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Dataset	11,839 articles · 9 Bengali news categories · mainstream journalism
Proposed model	LightGBM on multi-view TF-IDF (word + character + headline n-grams)
Headline result	84.21% accuracy and 0.842 F1-macro (ROC-AUC 0.924), beats published LSTM ensemble baseline (82.43%), $p = 0.012$
Core contribution	First LOCO benchmark , First N×N cross-category transfer matrix, Statistically validated improvement over prior work, SHAP-based explainability of transferable features, Reproducible evaluation protocol
Status	Code, experiments, and figures complete; Manuscript in preparation for submission to an IEEE conference.

Motivation

Existing Bengali fake news detectors are evaluated on random train/test splits, where articles from the same news category appear on both sides of the split. A model can score well by memorizing category-specific vocabulary rather than learning transferable deception cues — and that gap is invisible until the model meets a news category it has not seen before. This work introduces a Leave-One-Category-Out (LOCO) protocol that holds out an entire category at test time, directly simulating deployment on an unfamiliar news domain. The approach mirrors leave-one-domain-out evaluation used elsewhere for AI-generated text detection , where the same failure pattern — strong in-distribution accuracy that collapses on an unseen source — has been documented.

Approach

- Multi-view feature engineering: word TF-IDF (1–3 grams, 100K features), character TF-IDF (3–5 grams, 80K features, robust to Bengali morphological variation), and headline TF-IDF (1–3 grams, 30K features, headline text weighted 2× in the combined representation since deception signals concentrate there).
- Five classifiers benchmarked on the full feature matrix — Logistic Regression, Linear SVM, Random Forest, XGBoost, and LightGBM — against a previously published LSTM ensemble baseline (82.43% accuracy).
- LightGBM selected as the proposed model: highest standard-evaluation accuracy, strongest LOCO generalization, and efficient on sparse high-dimensional TF-IDF features, consistent with benchmarked trade-offs between gradient boosting variants reported in the broader literature.
- Dual-model LOCO protocol (Logistic Regression and LightGBM, both restricted to word-TF-IDF-only features to avoid category leakage) isolates whether generalization gaps come from the data itself or from model choice.
- A full N×9 source→target transfer matrix quantifies exactly how well a model trained on one category performs on every other category.
- SHAP TreeExplainer applied to the same word-TF-IDF LightGBM used in LOCO, so the explainability results interpret the features that cross-category generalization actually relies on — paired with a Jaccard overlap analysis of top features across categories.

Key Results

1. LightGBM beats the published LSTM baseline

On the standard (mixed-category) evaluation split, LightGBM reaches 84.21% accuracy and 0.842 F1-macro (ROC-AUC 0.924), a +1.78 percentage point improvement over the LSTM ensemble baseline (82.43%). A one-sided binomial test confirms the improvement is statistically significant ($p = 0.012$, 95% CI 82.8–85.6%). LightGBM also outperforms every other baseline tested — Logistic Regression (77.5%), Random Forest (76.7%), Linear SVM (79.9%), and XGBoost (81.8%).

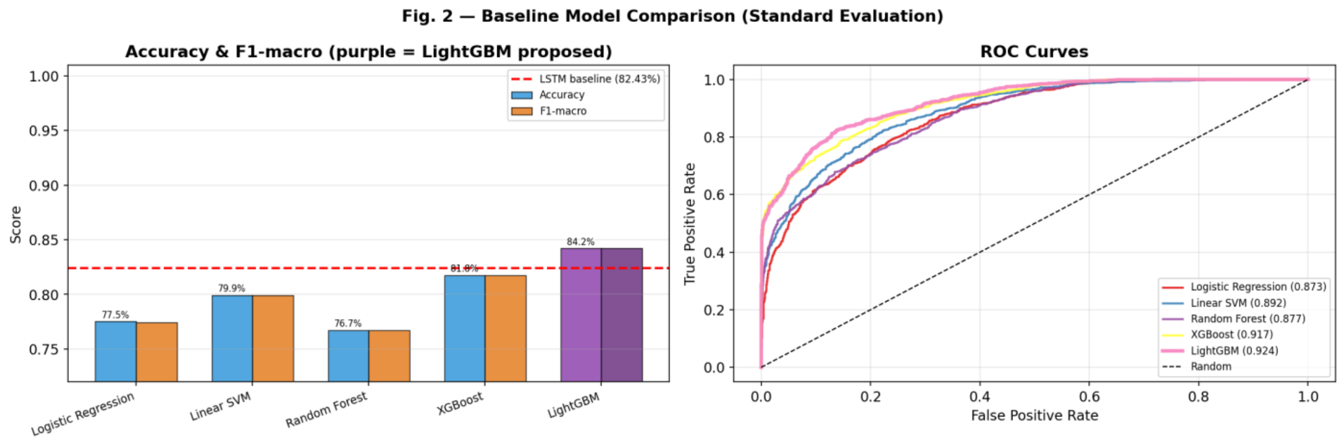

Fig. 1 — LightGBM vs. baseline classifiers on standard evaluation (left) and ROC curves (right).

2. Cross-category generalization is real, but uneven

Across the nine held-out categories, LightGBM's mean LOCO F1-macro is 0.712. Category difficulty varies substantially: Editorial is the hardest category to generalize to (LOCO F1 = 0.637), while Miscellaneous is the easiest (F1 = 0.797). A dual-model comparison shows LightGBM and Logistic Regression track each other closely across categories (Spearman $\rho = 0.717$, $p = 0.030$) — evidence that generalization difficulty is driven mainly by each category's own linguistic characteristics, not by which classifier is used.

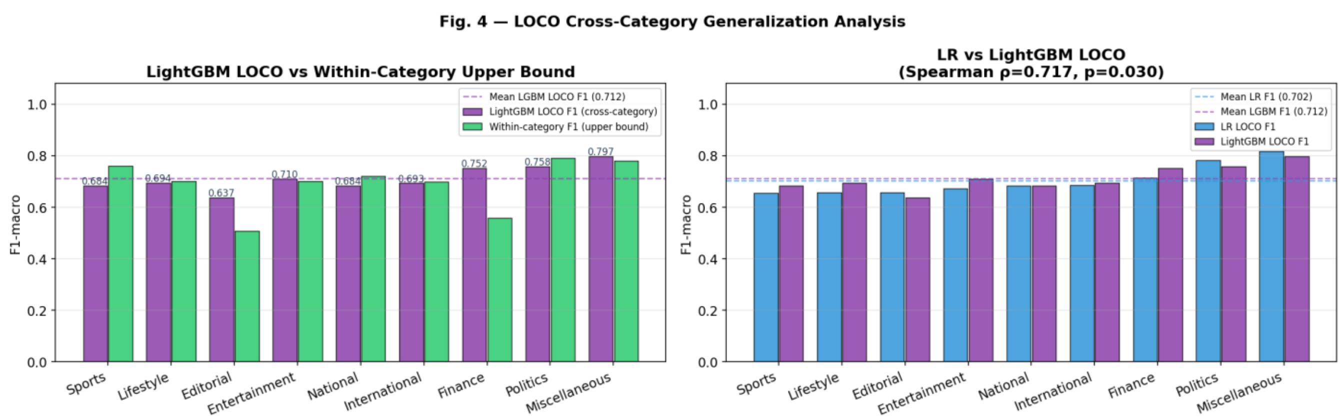

Fig. 2 — Per-category LOCO F1 vs. within-category upper bound (left); LightGBM vs. Logistic Regression LOCO performance by category (right).

3. Transfer cost is asymmetric and category-specific

The full N×N transfer matrix shows that cross-category transfer cost is not uniform: Politics is the most difficult source category to transfer out of (up to 0.41 F1 drop when tested on Sports), while transfer into Politics as a target is comparatively cheap. This asymmetry, combined with low Jaccard overlap among each category's top LightGBM features (mostly 0.05–0.33), indicates that categories rely on distinct, largely non-overlapping linguistic deception patterns — which is the underlying reason cross-category transfer is lossy.

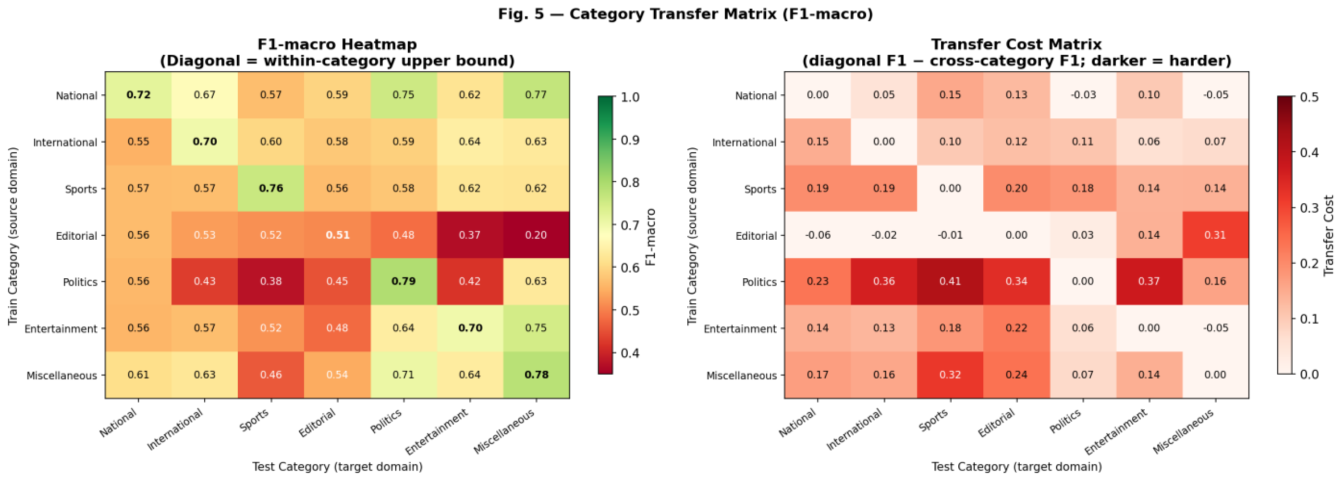

Fig. 3 — Category transfer matrix: F1-macro (left) and transfer cost, diagonal minus cross-category F1 (right).

Reproducibility

All experiments use a fixed random seed (42) throughout preprocessing, feature extraction, model training, and evaluation. No hyperparameter search was run on any model — the paper's contribution is the LOCO generalization protocol itself, not tuning, so well-validated default configurations are used for every classifier. Category one-hot encoding is deliberately excluded from LOCO training (used only in the standard evaluation) because including it would let the held-out category leak into the feature space.

Current Status

The codebase, all experiments (baseline comparison, LOCO evaluation, transfer matrix, SHAP explainability, statistical validation), and all figures above are complete. I am currently preparing the manuscript for submission to IEEE Conference.